

Supplemental Material for
“Amplitude-Robust Qubit Spectroscopy Using
Echo-Root-Lorentzian Pulse Shapes”

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This Supplemental Material provides the experimental setup, the Bloch-equation coherence limit, pulse-cutoff and pulse-duration analyses, the adiabatic-basis derivation, high-resolution and extended-amplitude scans, the numerical model, a comparison of the root-Lorentzian and echo-root-Lorentzian protocols across pulse parameters, and a comparison between simulation and experiment.

S1. MEASUREMENT SETUP

Experiments were performed on a QPU comprising 11 uncoupled superconducting qubits hosted in a dilution refrigerator. Control and readout were implemented using a Quantum Machines OPX platform together with microwave components from Keysight Technologies, including IQ mixers, local oscillators (LOs), attenuators, filters, and amplifiers.

A schematic of the setup is shown in Fig. S1. At room temperature, baseband I/Q waveforms are generated by arbitrary waveform generators (AWGs) and upconverted to microwave frequencies using IQ mixers driven by LOs. Separate lines are used for qubit drive and resonator readout. In addition, a dedicated DC bias line provides static flux control.

The microwave signals are routed into the dilution refrigerator through heavily attenuated and filtered coaxial lines. Attenuators are distributed across multiple temperature stages (300 K to ~ 9 mK) to thermalize the lines and suppress room-temperature noise. Additional filtering (low-pass and infrared filters) further reduces high-frequency noise before the signals reach the device.

The quantum device is mounted at the mixing chamber stage (~ 9 mK). The outgoing readout signal is routed through a chain of circulators that isolate the qubit from amplifier back-action, followed by cryogenic amplification using a high-electron-mobility transistor (HEMT) amplifier at the 4 K stage. The signal is then further amplified, downconverted, and digitized at room temperature.

Digitized signals are processed in real time and used for classical post-processing and iterative experiment execution.

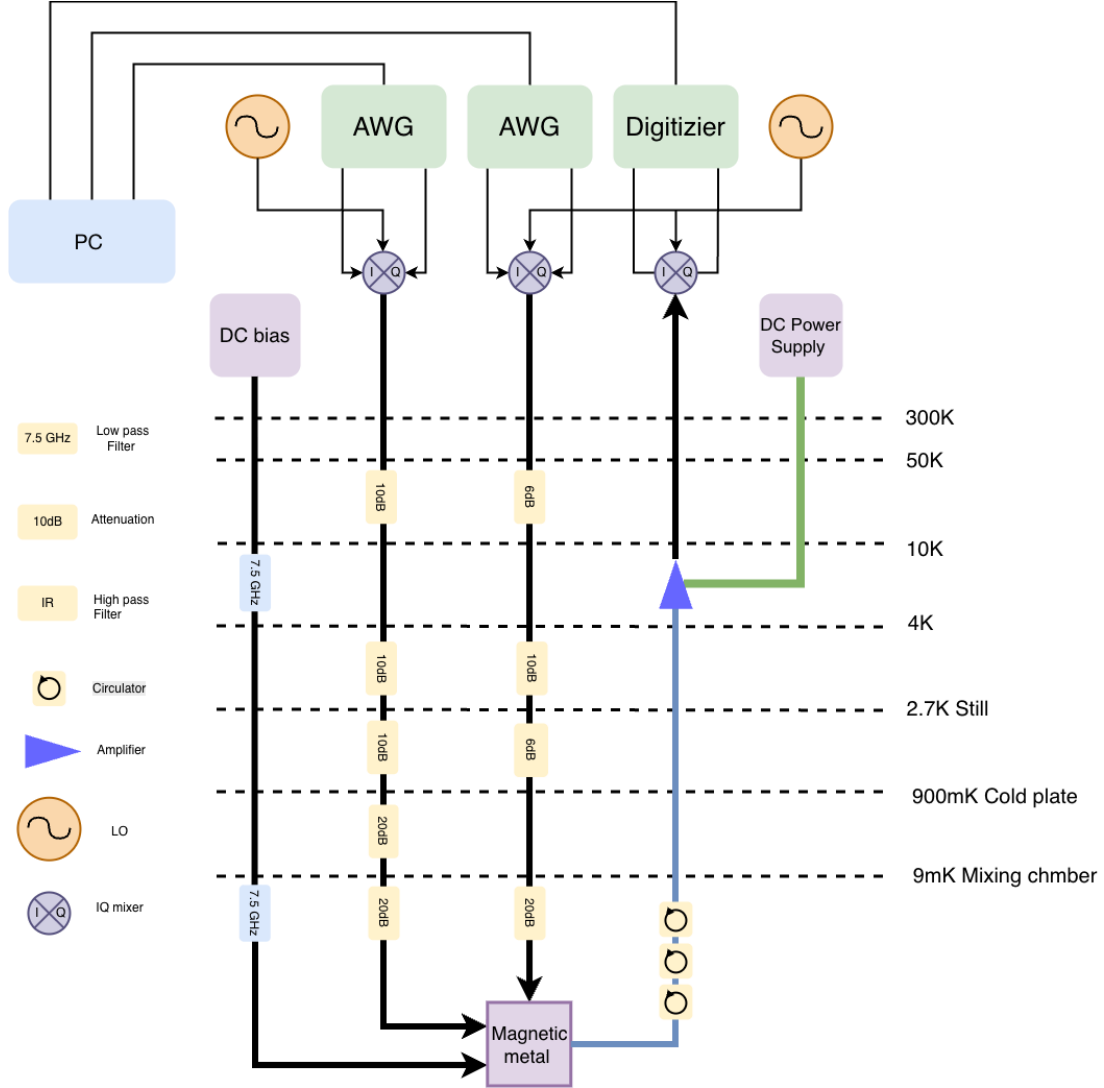


Figure S1. Schematic of the experimental setup for qubit control and readout. Room-temperature electronics (top) include a computer controlling two arbitrary waveform generators (AWGs) and a digitizer. The AWGs generate baseband I/Q signals that are upconverted to microwave frequencies via IQ mixers driven by local oscillators (LOs), enabling qubit drive and resonator readout. The signals are routed into a dilution refrigerator (dashed region), where multiple temperature stages (38 K, 10 K, 2.7 K, 900 mK, and 9 mK) are indicated. Input lines are attenuated and filtered at each stage to suppress thermal noise before reaching the device. The output signal is routed through circulators and amplified by a cryogenic high-electron-mobility transistor (HEMT) amplifier before being further processed at room temperature and digitized. The quantum device is mounted at the base temperature stage (9 mK).

S2. THE T_2 LIMIT

The Bloch equations [1] provide a theoretical framework for describing the dynamics of a two-level system under an external excitation field with strength Ω_0 , frequency detuned from natural qubit frequency with detuning Δ , including presence of relaxation and dephasing times T_1 and T_2 . Originally developed in the context of NMR, the Bloch equations are equally applicable to superconducting qubits.

The formalism is based on solving a system of differential equations for the expectation values of the qubit's Pauli operators, σ_x , σ_y and σ_z , which together give full information of the qubit state.

A key advantage of the Bloch solution is its validity across a broad range of excitation pulse durations, from short to long, and for both weak and strong driving fields (however, to prevent leakage into higher energy levels, the drive amplitude must remain below the qubit's anharmonicity).

In our study, we focus on the regime where the solution to the Bloch equations describes long-duration pulses, such that the qubit undergoes complete energy saturation $t \gg T_1$. In this limit, for resonance pulses, the qubit reaches a mixed state, and its evolution is a simple closed-form solution. In particular, we are interested in the expectation value $\langle \sigma_z \rangle$, which serves as the measurable observable of our system:

$$\langle \sigma_z \rangle = \frac{1 + (T_2 \Delta)^2}{1 + (T_2 \Delta)^2 + \Omega_0^2 T_1 T_2}. \quad (\text{S1})$$

The solution determines the linewidth at broad range of cases and will be applicable to understand the limits for high resolution spectroscopy.

S2.1. Linewidths and Saturation Spectroscopy

In the strong-driving regime, where the Rabi frequency dominates the detuning term,

$$\Omega_0 \gg \sqrt{\frac{T_2}{T_1}} \Delta,$$

Eq. (S1) simplifies to

$$\langle \sigma_z \rangle \approx 1 - 2 \frac{1}{1 + \frac{\Delta^2}{(T_1/T_2)\Omega_0^2}}. \quad (\text{S2})$$

In this limit, the transition linewidth is broadened to

$$\Delta_{1/2,\Omega_0} = \frac{2\Omega_0}{\sqrt{T_2/T_1}}. \quad (\text{S3})$$

This phenomenon, known as *power broadening* (or saturation broadening), is analogous to the linewidth increase observed in Rabi spectroscopy. Here, however, the broadening arises from saturation of the transition and the resulting population redistribution between the qubit states.

In contrast, in the weak-driving regime,

$$\Omega_0 \ll \sqrt{\frac{T_2}{T_1}} \Delta,$$

the excitation remains far from saturation, and the linewidth is determined solely by the qubit coherence time. In this case, Eq. (S1) reduces to

$$\langle \sigma_z \rangle \approx 1 - 2\Omega_0^2 T_1 T_2 \frac{1}{1 + T_2^2 \Delta^2}, \quad (\text{S4})$$

which has a Lorentzian lineshape with a minimum achievable linewidth set by T_2 :

$$\Delta_{1/2,T_2} = \frac{2}{T_2}. \quad (\text{S5})$$

This represents the coherence-limited linewidth of the qubit transition and constitutes the fundamental resolution limit of conventional continuous-wave spectroscopy.

S3. PULSE CUTOFF

The spectral linewidths corresponding to the two regimes are given by

$$\Delta_{1/2,T_2} = \frac{1}{\pi T_2}, \quad \Delta_{1/2,\Omega_0} = \frac{\Omega_c}{\pi \sqrt{T_2/T_1}}.$$

Requiring $\Delta_{1/2,T_2} < \Delta_{1/2,\Omega_0}$ yields

$$\Omega_c < \frac{1}{\sqrt{T_1 T_2}}.$$

If we express the coupling strength as $\Omega_c = c \Omega_0$, the condition becomes

$$c < \frac{1}{\Omega_0 \sqrt{T_1 T_2}}.$$

Substituting $T_1 = 30 \mu\text{s}$, $T_2 = 15 \mu\text{s}$, and $\Omega_0 = 50 \text{ MHz}$ gives

$$c < 9.4 \times 10^{-4},$$

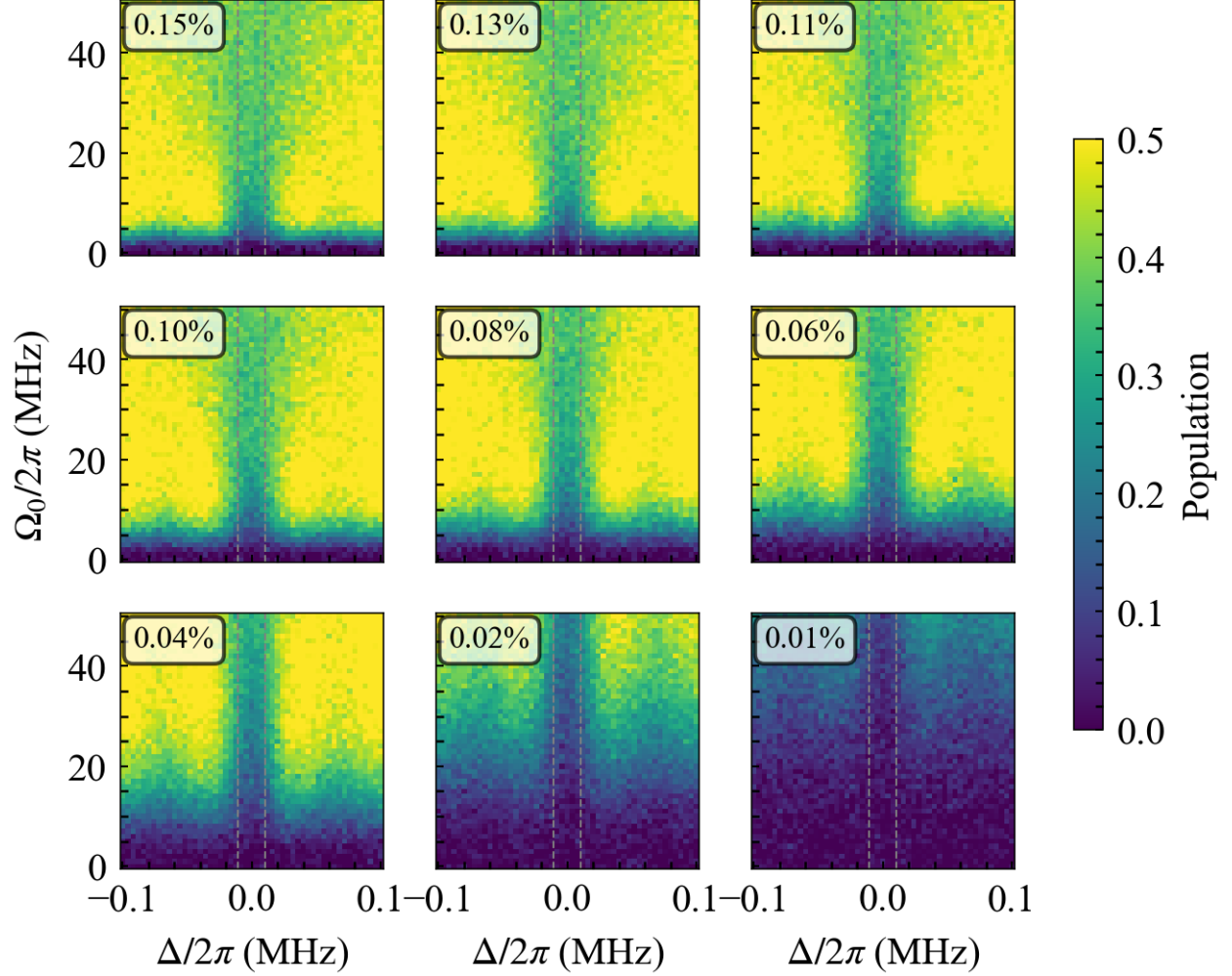


Figure S2. Spectroscopic response obtained using echo-root-Lorentzian pulses for different relative cutoff amplitudes (indicated in the upper-left corner of each panel). Increasing the cutoff smooths the phase inversion point of the pulse, thereby suppressing derivative discontinuities that otherwise induce residual power broadening. As a result, the spectral line becomes progressively narrower and more robust to drive-amplitude variations. However, for excessively small cutoff values the effective signal-to-noise ratio deteriorates, limiting the attainable spectral contrast despite reduced broadening.

S4. ADIABATIC BASIS

The matrix form of the Hamiltonian in the rotated frame takes the form:

$$H(t) = \frac{1}{2} \begin{bmatrix} -\Delta & \Omega(t) \\ \Omega(t) & \Delta \end{bmatrix}. \quad (\text{S6})$$

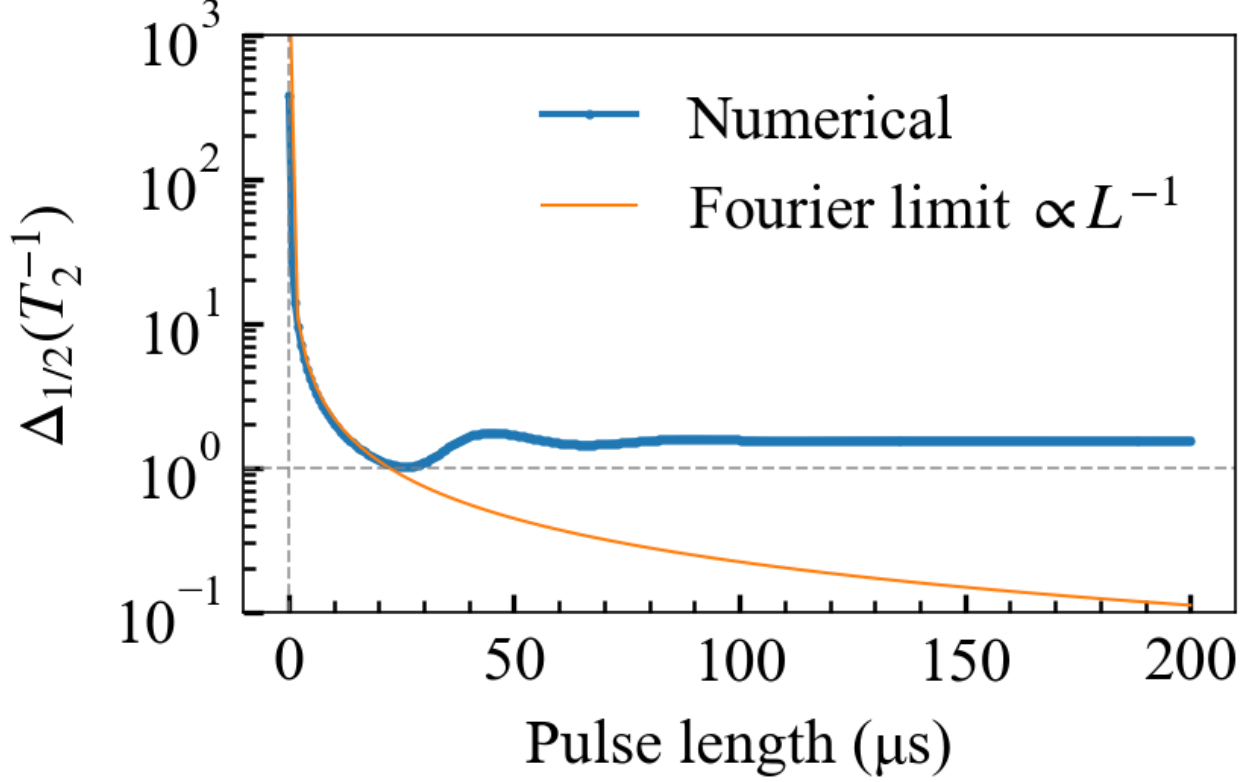


Figure S3. Numerical linewidth $\Delta_{1/2}$, normalized to the decoherence-limited scale T_2^{-1} , obtained from square-pulse spectroscopy as a function of pulse duration. For short interrogation times, the linewidth decreases inversely with pulse length, following the Fourier limit. For longer pulses, the linewidth saturates at the fundamental T_2 limit imposed by decoherence, deviating from the Fourier-limited scaling.

The Rabi frequency $\Omega(t)$ quantifies the field-induced coupling between the two states. The instantaneous eigenstates of $H(t)$, commonly referred to as the adiabatic states, are:

$$|w_-(t)\rangle = \cos \vartheta(t)|0\rangle - \sin \vartheta(t)|1\rangle, \quad (\text{S7})$$

$$|w_+(t)\rangle = \sin \vartheta(t)|0\rangle + \cos \vartheta(t)|1\rangle, \quad (\text{S8})$$

where the mixing angle is given by $\vartheta(t) = \frac{1}{2} \arctan \frac{\Omega(t)}{\Delta}$. The corresponding eigenvalues, representing the adiabatic energy levels, are:

$$\varepsilon_{\pm}(t) = \frac{\hbar}{2} \left(\Delta \pm \sqrt{\Omega^2(t) + \Delta^2} \right), \quad (\text{S9})$$

yielding an energy gap of $\varepsilon(t) = \varepsilon_+(t) - \varepsilon_-(t) = \hbar \sqrt{\Omega^2(t) + \Delta^2}$.

Using the eigenvectors from Eq. (S8) we define the adiabatic transformation matrix R .

Applying this transformation to the Hamiltonian in Eq. (S6) gives the Hamiltonian in the adiabatic basis $H^a = R^\dagger H R - i R^\dagger \dot{R}$, which can be explicitly written as:

$$H^a(t) = \frac{1}{2} \begin{bmatrix} -\varepsilon(t)/2 & -i\dot{\vartheta}(t) \\ \dot{\vartheta}(t) & \varepsilon(t)/2 \end{bmatrix}. \quad (\text{S10})$$

Adiabatic evolution requires that the rate of change of the mixing angle $\vartheta(t)$ is much smaller than the energy splitting $\varepsilon(t)$, leading to the adiabatic condition [2]:

$$|2\dot{\vartheta}| \ll \varepsilon(t), \quad (\text{S11})$$

where the non-adiabatic coupling term is given explicitly by $|\dot{\vartheta}| = \frac{\Delta \dot{\Omega}(t)}{2[\Delta + \Omega(t)]}$.

Replacing the inequality in Eq. (S11) with an equality provides a approximate boundary condition between adiabatic and non-adiabatic regions

$$\sqrt{\Omega^2(t_0) + \Delta_b^2} = \frac{\Delta_b \dot{\Omega}(t_0)}{2[\Delta^2 + \Omega^2(t_0)]}, \quad (\text{S12})$$

where t_0 denotes the time at which the non-adiabatic coupling $|\dot{\vartheta}|$, is maximal, and Δ_b marks the boundary between the adiabatic and non-adiabatic regimes. This condition allows us to extract a threshold value for the detuning range $\Delta < \Delta_b$ which marks the range of detunings where post-pulse excitation remains significant. The excitational linewidth scales proportionally to the threshold detuning $\text{FWHM} \propto \Delta_b$. Consequently, we are particularly interested in the functional dependence of Δ_b on the peak Rabi frequency Ω_0 .

S5. PULSE LENGTH

The pulse duration imposes an additional constraint on spectroscopic resolution, as the spectral width of the drive in the frequency domain scales inversely with the pulse length. Figure 8 presents a numerical calculation of the extracted linewidth obtained using a square pulse, showing that for short pulse durations ($\leq 20 \mu s$), the measured spectroscopic linewidth is dominantly limited by Fourier broadening rather than decoherence.

In contrast, for pulse durations exceeding approximately $20 \mu s$, the echo-based pulse sequence approaches the decoherence-limited regime, yielding spectroscopy with an $\mathcal{O}(1)$ agreement with the T_2 -limited linewidth.

For longer pulse durations, an additional effect emerges: the off-resonant background signal gradually decays while the resonance feature itself remains approximately constant.

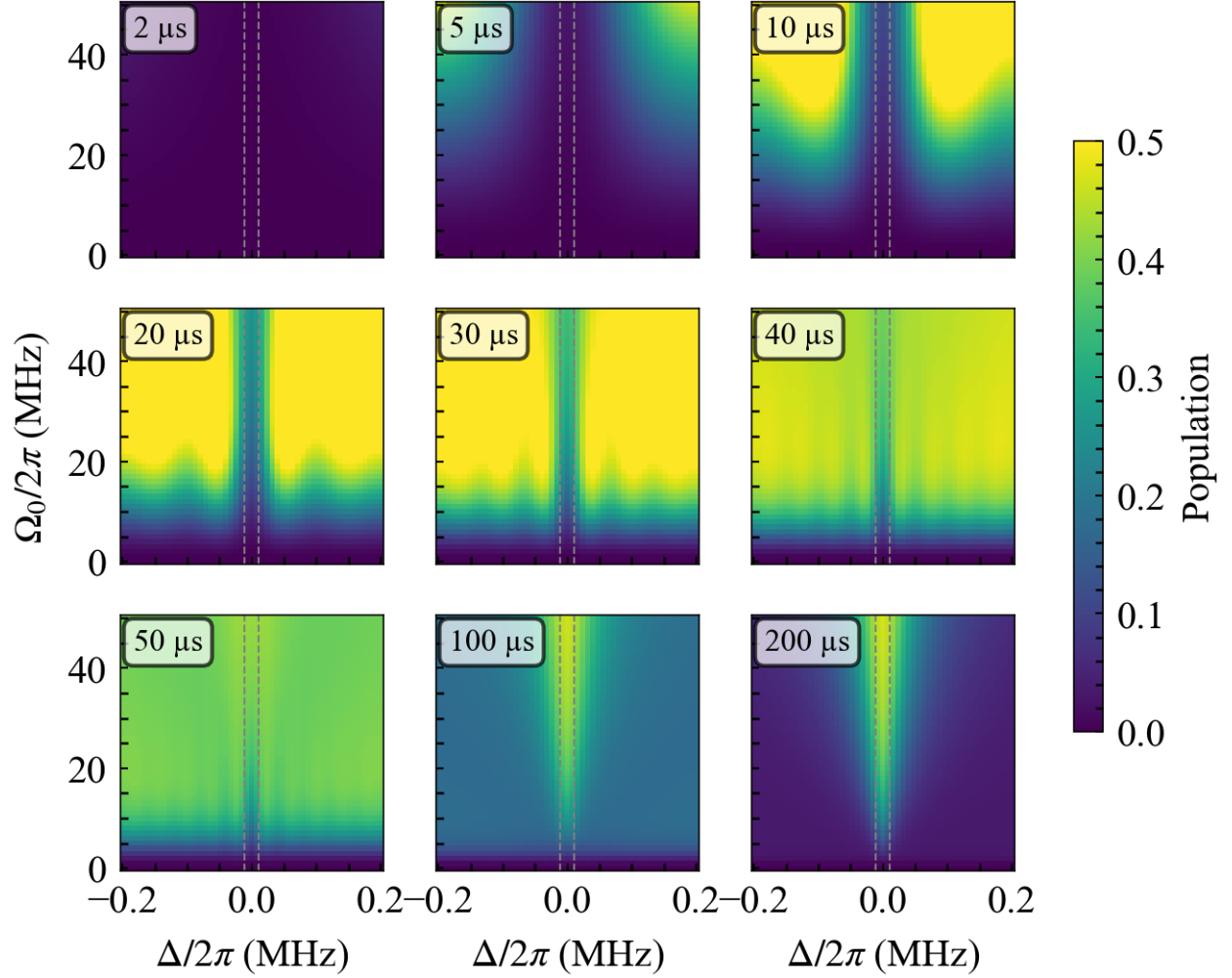


Figure S4. Numerical spectroscopic response obtained using echo-root-Lorentzian pulses for different pulse durations (indicated in each panel). As the pulse length increases, the spectral linewidth progressively narrows, approaching the decoherence-limited resolution set by the coherence time T_2 . For pulse durations comparable to or exceeding the relaxation time T_1 , population decay during the interrogation leads to an elevated off-resonant background, reducing spectral contrast despite the continued narrowing of the resonance feature.

This results in the formation of a characteristic narrow peak nested within the broader echo-induced dip, reflecting the onset of adiabatic state evolution in the vicinity of the resonance.

S6. ULTRA-HIGH-RESOLUTION SPECTROSCOPY AT LARGE DRIVE AMPLITUDES

S6.1. Broad and Narrow Frequency Scans

High-resolution spectroscopy at large drive amplitudes is characterized on two frequency scales. A broad scan first locates the qubit transition and reveals the surrounding baseline, nearby spectral features, and any strong-drive distortions. A narrow scan centered on the transition then samples the sharp echo-root-Lorentzian depletion feature densely enough to determine its center and linewidth. Presenting the broad and narrow scans together distinguishes a genuinely narrow resonance from a local feature embedded in a wider distorted response and demonstrates that the high resolution persists within the full spectral landscape.

S6.2. Extended-Amplitude Regime and Strong-Drive Distortions

At drive amplitudes above approximately 50 MHz, the Rabi frequency approaches half the qubit anharmonicity and the measured response begins to depart from an effective two-level description. In this extended-amplitude regime, the proximity of the first two-photon transition provides a natural interpretation of the additional spectral structure; a quantitative assignment would, however, require a converged multilevel model.

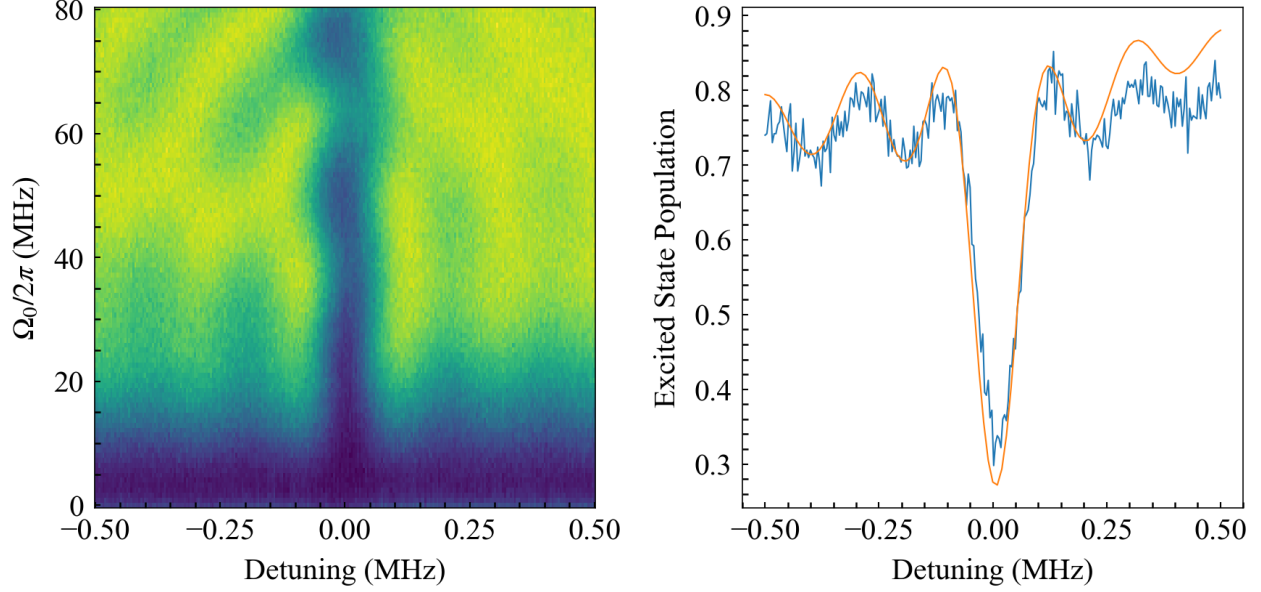


Figure S5. Two-dimensional spectroscopy map obtained with a $10\ \mu\text{s}$ echo-root-Lorentzian pulse at extended drive amplitudes (up to 80 MHz). The left panel shows the full detuning–amplitude map, while the right panel presents a line cut at fixed drive strength (blue) together with a smooth fit (orange). The observed distortions mark the breakdown of the ideal effective two-level response as the drive approaches the multilevel regime.

S7. NUMERICAL MODEL

The simulations use an effective driven two-level system coupled to Markovian relaxation and dephasing channels. In the frame rotating at the drive frequency, and within the rotating-wave approximation, the Hamiltonian is implemented as

$$\frac{H(t)}{\hbar} = \Delta a^\dagger a + \frac{\Omega_0}{2} f(t) (a + a^\dagger), \quad (\text{S13})$$

where $\Delta = \omega_q - \omega_d$ is the qubit–drive detuning, Ω_0 is the peak Rabi frequency, and $a = |0\rangle\langle 1|$ is the lowering operator. Up to an irrelevant identity term and the chosen sign convention for Δ , Eq. (S13) is equivalent to $H/\hbar = (\Delta/2)\sigma_z + (\Omega_0 f/2)\sigma_x$.

For a root-Lorentzian pulse of total duration L , the dimensionless envelope is

$$f_{\text{RL}}(t) = \left[1 + \left(\frac{t}{\tau} \right)^2 \right]^{-1/2}, \quad -\frac{L}{2} \leq t \leq \frac{L}{2}, \quad (\text{S14})$$

with τ chosen such that $f_{\text{RL}}(\pm L/2) = c$, where c is the relative cutoff. The echo variant is obtained by changing the drive phase by π at the pulse center,

$$f_{\text{ERL}}(t) = \text{sgn}(-t)f_{\text{RL}}(t). \quad (\text{S15})$$

The overall sign in Eq. (S15) is a phase convention and does not affect the measured populations.

The density matrix evolves according to the Lindblad master equation

$$\dot{\rho} = -\frac{i}{\hbar}[H(t), \rho] + \mathcal{D}\left[\sqrt{\frac{1}{T_1}} a\right] \rho + \mathcal{D}\left[\sqrt{\frac{2}{T_\phi}} a^\dagger a\right] \rho, \quad (\text{S16})$$

where $\mathcal{D}[C]\rho = C\rho C^\dagger - \frac{1}{2}\{C^\dagger C, \rho\}$ and

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}. \quad (\text{S17})$$

Each calculation starts in $|0\rangle$, integrates Eq. (S16) over the complete pulse, and records the final Bloch-vector components. Spectroscopy maps are generated by repeating this evolution over the experimental detuning and drive-amplitude grids, using the pulse duration, cutoff, T_1 , and T_2 associated with the corresponding data set.

The numerical Hilbert space is restricted to two levels. Consequently, this model captures coherent drive dynamics, relaxation, and pure dephasing, but it does not describe leakage, AC-Stark shifts produced by higher transmon levels, or multiphoton transitions. Comparisons in the extended-amplitude regime are therefore interpreted as tests of where the effective two-level description breaks down rather than as a simulation of those multilevel effects.

S8. COMPARISON OF ROOT-LORENTZIAN AND ECHO-ROOT-LORENTZIAN PULSES

The two protocols are compared over the principal experimental control parameters: pulse duration, relative cutoff, and peak drive amplitude. For each parameter set, we examine the final spectroscopic response, extracted linewidth, signal contrast, and linewidth-to-signal ratio. Pulse-length sweeps distinguish the Fourier-limited and decoherence-limited regimes, while cutoff sweeps quantify broadening introduced by truncating the formally infinite Lorentzian tail. Drive-amplitude sweeps then test whether the narrow feature remains

stable as the Rabi frequency is increased. These comparisons show where the phase inversion of the echo-root-Lorentzian pulse suppresses coherent oscillations and produces a broader, more uniform operating region than the corresponding root-Lorentzian pulse.

S8.1. Dependence on Pulse Length

At fixed cutoff and peak Rabi frequency, varying the pulse length separates short-pulse Fourier broadening from the long-pulse limit imposed by decoherence. The comparison also reveals amplitude-dependent coherent oscillations in the root-Lorentzian response and their suppression by the echo phase inversion.

S8.2. Dependence on Pulse Cutoff

At fixed duration and peak Rabi frequency, varying the relative cutoff tests the sensitivity of both protocols to truncation of the Lorentzian tail. The relevant scale is the residual endpoint drive, $c\Omega_0$, which must remain small compared with the desired linewidth scale.

S8.3. Dependence on Drive Amplitude

Finally, amplitude sweeps compare the linewidth and contrast of the two pulse shapes over their full useful operating ranges. The echo protocol is assessed by the continuity and uniformity of its narrow-linewidth region, in addition to its minimum attainable linewidth.

S9. COMPARISON BETWEEN SIMULATION AND EXPERIMENT

Simulation and experiment are compared using the same final-state spectroscopic observable and the same linewidth-extraction procedure. The comparison tests the resonance center, linewidth, and contrast across drive amplitude, with the measured T_1 and T_2 supplied to the numerical model. Agreement within the two-level operating range supports the proposed echo-cancellation mechanism. Systematic deviations at the largest amplitudes identify the onset of physics omitted from the effective model, including higher-level coupling and strong-drive calibration effects.

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- [1] F. Bloch, Nuclear induction, Phys. Rev. **70**, 460 (1946).
- [2] I. I. Boradjiev and N. V. Vitanov, Power narrowing in coherent atomic excitation by smoothly shaped pulsed fields, Optics Communications **288**, 91 (2013).